Higher National Diploma in Quantity Surveying

Second Year, First Semester Examination – 2020

QS21023- Building Economics I

Instructions for Candidates:
Answer any **Five (05)** questions.
All question carry equal marks

No. of questions : 06
No. of pages : 03
Time : Three (03) hours

Question 01

i) "Economics is a social science" Briefly explain above statement? **(03 Marks)**

ii) What are the three basic problems in economics? **(03 Marks)**

iii) What are the four production resources? **(03 Marks)**

iv) What is opportunity cost? Explain with example. **(03 Marks)**

v) What is the "Marginal Utility"? Briefly explain the relationship between Marginal Utility and total utility. **(03 Marks)**

vi) What are the factors determining Demand? **(02 Marks)**

vii) What is the meaning of Division of Labour? Briefly explain the merits and demerits of the division of labour? **(03 Marks)**

(Total 20 Marks)

Question 02

RIBA/SLIA Plan of work is the stages of proceeding from feasibility stage to completion stage of the project

i. Explain why it is important to follow the RIBA/SLIA plan of work for formulation of Project? **(04 Marks)**

ii. Name four feasibilities which have been considered in Project Appraisal stage. Briefly explain two of them? **(04 Marks)**

iii. What is the RIBA/SLIA plan of work stages of Quantity Surveyor involved under the Pre contract and Post Contract stage? **(04 Marks)**

iv. What are the stages of RIBA/SLIA related to Pre Contract stage? Explain the stages of Project Appraisal and conceptual design? **(04 Marks)**

v. What do you mean by EIA report for a project? Briefly explain the content of EIA report and who are the professionals involved to prepare EIA report. **(04 Marks)**

(Total 20 Marks)

Question 03

i. What do you understand by the term **"supply"** in economics? **(03 Marks)**

ii. What is the meant by the term **"equilibrium"** in a market structure? **(03 Marks)**

iii. Who are the key **stakeholders** in construction industry? **Name at least three** stakeholders in the construction industry. **(04 Marks)**

iv. Name the four Regulator bodies and their roles in Construction Industry. **(04 Marks)**

v. Explain briefly main categories of civil engineering contractors graded by CIDA for the registration? **(03 Marks)**

vi. If a Contractor required registering his company for Civil Engineering Contracts in CIDA, what are the qualifications the Contractor should provide to CIDA? **(03 Marks)**

(Total 20 Marks)

Question 04

i. Explain the different kind of estimates and when they are applied. **(03Marks)**

ii. You are required to submit a rough estimate for Warehouse construction of 6m high 4000 square meter in 2020. Cost data and cost indices relative to 2013 and 2020 are given in table below with relative to the civil construction. Estimate following

a. Unit cost per cubic meter in warehouse constructed in year 2013 **(02 Marks)**

b. Unit cost per cubic meter in warehouse proposed to be constructed in 2020? **(03 Marks)**

c. Total cost of proposed warehouse to be constructed in 2020 **(02 Marks)**

Year	Floor Area in sqm	Height in m	Cost in Rs.	Unit Cost per cum	Construction Cost index
2013	3000	9	3,500,000.00		130
2020	4000	7			175

iii. Domestic two story Building with floor area in ground floor is 200 sqm and first floor 120 sqm. Unit rate per sqm is Rs 60,000/= for ground floor and 70,000/= for first floor. Reduction of floor area for Stairs and Courtyard is 10% for ground floor and 5% for first floor respect to total area of ground floor and first floor. Following cost are related as percentage of Preliminary cost Estimates.

Architectural and structural design cost is 3%.

Utility service cost (Water, Electricity and Sanitary) cost is 9%

Construction supervision cost is 4%

a) Calculate the effective plinth area for Ground floor

b) Calculate the effective plinth area for First floor
c) What is the plinth area cost for Ground floor?
d) What is the plinth area cost for First Floor?
e) What is the Total Plinth area Cost for Building?
f) Estimate the Architectural and Design cost
g) Estimate the Utility Service Cost
h) Calculate the total estimate cost of Building
i) What will be the accuracy of preliminary estimate compare with detail Estimate (BOQ)?
j) Why Rough or Preliminary estimate is important in Client point of view?

(10 x 1 = 10 Marks)

(Total 20 Marks)

Question 05

i. Briefly explain the following economic systems,
   a. Traditional Economy
   b. Capitalist Economy
   c. Socialist Economy
   d. Mixed Economy **(08 Marks)**

ii. What are the merits and demerits of a mixed economy? **(03 Marks)**

iii. What do you mean by "Consumer's surplus"? **(03 Marks)**

iv. What do you mean by the term "production"? **(03 Marks)**

v. List down the **three** types of **products** which are related to **the production function**? **(03 Marks)**

(Total 20 Marks)

Question 06

i. Explain the law of Demand? **(04 Marks)**

ii. Answer the following
   a. What are the three types of elasticity of demand? **(03 Marks)**
   b. Calculate price elasticity of demand by using following details and mention that demand is elastic or inelastic. When price of X rises by 10%, demand for X falls by 5% **(04 Marks)**

iii. What is elasticity of supply? Mention the factors determining elasticity of supply? **(05 Marks)**

iv. Briefly explain the **law of diminishing marginal utility**? **(04 Marks)**

(Total 20 Marks)